Physical Chemistry CUME

Quantum Mechanics

(Particle in a box and basic understanding of quantum concepts such as observable, operators, Schrödinger's equation.)

1. Consider a particle in a box with length L (one dimension), the center of the box is the origin of the x axis. The ground state solution is $\psi(x) = \mathrm{N}\cos(\frac{\pi x}{L})$. Please calculate the normalization constant N. [show your work] (20 points)

2. What is the total energy of the ground state for the system described above? What is the expectation value of the kinetic energy for this state? What is the expectation value of the potential energy for this state? (30 points)

3. The ground state of particle in a box can be understood as a linear combination of two travelling waves, one in the positive x direction, $\varphi_r(x) = e^{ikx}$; and one in the negative x direction, $\varphi_l(x) = e^{-ikx}$, where k is frequently referred to as the wave-number (radian per meter). One way to understand this is with the Euler identity $\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$. Please express the ground state wave-function as a linear combination of two traveling waves. Please write down the expression for *k* to receive full credit. (20 points)

4. What is the average momentum square $\langle \hat{p}^2 \rangle$ for the ground state of particle in a box? What is the average momentum $\langle \hat{p} \rangle$? Please calculate the momentum of each of the two traveling waves. [All the answers in 4 must be expressed as multiples of *k* to receive full credit unless the answer is 0.] (30 points)

Integrals you may or may not need: $\int \cos(ax)dx = \frac{\sin(ax)}{a}$, $\int \cos^2(ax)dx = \frac{x}{2} + \frac{\sin(2ax)}{4a}$.